QPCODE **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

4th semester MGU-UGP(Honours) Examination

DSE- MG4DSECFS201-LINUX ADMINISTRATION (Operating System Architecture Specialization)

(2024 ADMISSION ONWARDS)

Duration: 2Hrs **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **10** questions

Each question carries **2** marks

1. State any two functions of a text editor. [K] /[CO1]
2. Explain the purpose of the *finger* command in Linux [U] /[CO1]
3. List any two types of files in Linux [K] /[CO1]
4. Explain the basic hardware requirements for installing Linux [U] /[CO1]
5. Illustrate the use of the *uniq* command with a simple example. [A] /[CO2]
6. Explain the function of the *kill* command in Linux. [U] /[CO2]
7. Demonstrate the use of the *expr* command to perform a simple arithmetic operation. [U] /[CO2]
8. Differentiate between global variables and local variables in shell programming [An] /[CO3/CO4]
9. Explain the purpose of shell variables. [U] /[CO3/CO4]
10. Illustrate the method of declaring and initializing variables in shell programming. [U] /[CO3/CO4]
11. Name the command used to schedule tasks in Linux and Explain [K] /[CO3/CO4]
12. Explain the purpose of RPM in Linux. [U] /[CO3/CO4]

[2x10 = 20]

Part B

Short Essay Type Questions

Answer any 6 questions

Each question carries 5 marks

13. Discuss the features of Linux with suitable examples. [U] /[CO1]
14. Explain the Linux commands used for managing directories with examples.[U] /[CO1]
15. Identify the mathematical commands used in Linux with suitable examples[U] /[CO2]
16. Compare the *mount* and *umount* commands in Linux [A] /[CO2]
17. Illustrate a shell program in Linux to find the sum of digits of a number.[A] /[CO3/CO4]
18. Demonstrate a shell script using if to calculate the grade of a student based on given marks and display the appropriate grade. [An] /[CO3/CO4]
19. Analyse the duties of a System Administrator by examining their role in system maintenance, security management, and user support in a Linux environment. [An] /[CO3/CO4]
20. Differentiate Linux user management commands with respect to user creation, modification, and deletion. [An] /[CO3/CO4]

[5x6 = 30]

Part C

Essay Type Questions

Answer any **2** questions

Each question carries **10** marks

21. Explain the concept of simple filter commands in Linux. [U] /[CO2]
22. Analyse the different control structures used in Linux shell scripting and explain how decision-making and looping structures control the flow of execution. An] /[CO3/CO4]
23. Compare DHCP, DNS, Apache and Squid. [An] /[CO3/CO4]